

# A Closed Form for the Hodgson-Kerckhoff Tube-Packing Coefficient — updated 2026-05-04

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A CLOSED FORM FOR THE HODGSON-KERCKHOFF TUBE-PACKING COEFFICIENT

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D. B. — April 2026 (updated 4 May 2026)

Gap Geometry — <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

OSF: [osf.io/zx4g7](https://osf.io/zx4g7)

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★ UPDATE NOTICE — 4 May 2026 ★

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Section 3 has been corrected and a new Section 3.1 added.

## (1) §3 ELLIPSE AREA CORRECTION

The earlier text contained the line:

"At  $R_0$ , the ellipse area simplifies completely:  
 $\pi * \sinh^2(R_0) * \cosh(2R_0) = \pi * (1/2) * 2 = \pi$   
Therefore  $A_e = \sqrt{2} * \ln(2) * \pi$  at the critical radius."

This was a misderivation. Hodgson-Kerckhoff's actual ellipse area formula (HK 2002 Theorem 4.4 p. 29; HK 2007 Theorem 5.6 p. 36) is:

$$A_e = \pi * \sinh^2(R_{\text{hat}}) / (2 * S * \cosh(2 * R_{\text{hat}})) \\ = \sqrt{2} * \ln(2) * \pi * \sinh^2(R_{\text{hat}}) / (2 * \cosh(2 * R_{\text{hat}}))$$

since  $1/S = \sqrt{2} * \ln(2)$ . At  $R_0$  this gives  $\sqrt{2} * \ln(2) * \pi / 8$ , not  $\sqrt{2} * \ln(2) * \pi$ . The conclusion in the earlier text was off by a factor of 8.

The error originated in a transcription of HK 2007 that dropped the factor of 2 in the denominator and was carried into the rewrite. Both HK papers (math/0204345v1.pdf and 0709.3566v1.pdf) confirm the corrected formula independently.

§3 has been rewritten with HK's actual formula and the  $\sqrt{2} * \ln(2) * \pi / 8$  conclusion. The structural argument and the central closed-form identity  $1/S = \sqrt{2} * \ln(2)$  are unchanged and unaffected.

## (2) §3.1 NEW — SECOND APPEARANCE OF $\sqrt{2} * \ln(2)$ IN HK 2007

A new subsection §3.1 documents that the same numerical constant 0.980258 also appears as the coefficient of the Euclidean injectivity radius lower bound  $c(R) = 0.980258 / (\coth R + 1)$  on page 41 of HK 2007. Two independent contexts in the same paper, both unrecognized as  $\sqrt{2} * \ln(2)$  for nineteen years.

## ARCHIVED PRIOR VERSION

Prior published version archived as  
OLD-2026-05-04\_HK\_Closed\_Form\_SHORT\_DRAFT.txt and  
OLD-2026-05-04\_HK\_Closed\_Form\_SHORT.pdf in OLD/ subfolder.

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ABSTRACT

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The numerical constant 0.980258 appearing in the tube-packing bound of Hodgson & Kerckhoff (Annals of Mathematics, 2005) is shown to equal  $\sqrt{2} * \ln(2)$  exactly. The identity follows from one algebraic simplification –  $\operatorname{arcsinh}(1/(2*\sqrt{2})) = \ln(2)/2$  – applied to the authors' own proof of Theorem 4.4. This closed form does not appear in the original paper or in any subsequent citation as of April 2026.

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## 1. THE COEFFICIENT

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### SOURCE:

C. D. Hodgson & S. P. Kerckhoff  
 "Universal bounds for hyperbolic Dehn surgery"  
 Annals of Mathematics, 162(1), 367-421, 2005  
 Preprint: arXiv:math/0204345 (April 2002)

In the proof of Theorem 4.4 (page 29, arXiv numbering), the authors bound  $|\sinh(\zeta)/\zeta|$  over the relevant range by defining:

$$S = (1/(2*\sqrt{2})) / \operatorname{arcsinh}(1/(2*\sqrt{2}))$$

They compute  $S \sim 1/0.980258$ , so  $1/S \sim 0.980258$ .

This coefficient controls the semi-major axis of embedded ellipses in the tube-packing argument that underlies all subsequent bounds in the paper, including:

- The area bound:  $\operatorname{area}(T_R) \geq 3.3957 * \sinh^2(R) / \cosh(2R)$
- The packing function  $h(r) = 3.3957 * \tanh(r) / \cosh(2r)$
- The normalized-length threshold (7.515, refined to 7.5832 in [2])
- The bound on non-hyperbolic Dehn fillings (at most 60)

The constant  $3.3957 = 2*\sqrt{3}/S = 2*\sqrt{3} * (1/S)$ .

### CONTINUED USE:

The coefficient remains in active use without alteration in:

D. Futer, J. S. Purcell, S. Schleimer  
 "Effective bilipschitz bounds on drilling and filling"  
 Geometry & Topology, 26(3), 1077-1188, 2022

D. Futer, J. S. Purcell, S. Schleimer  
 "Effective drilling and filling of tame hyperbolic 3-manifolds"  
 Commentarii Mathematici Helvetici, 97(3), 457-512, 2022

No closed form for 0.980258 appears in any subsequent citation.

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## 2. THE CLOSED FORM

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### THE KEY IDENTITY:

$$\operatorname{arcsinh}(x) = \ln(x + \sqrt{x^2 + 1})$$

Let  $x = 1/(2*\sqrt{2})$ :

$$\begin{aligned} x^2 &= 1/8 \\ x^2 + 1 &= 9/8 \\ \sqrt{x^2 + 1} &= 3/(2*\sqrt{2}) \end{aligned}$$

$$x + \sqrt{x^2 + 1} = 1/(2*\sqrt{2}) + 3/(2*\sqrt{2})$$

$$\begin{aligned}
&= 4/(2*\sqrt{2}) \\
&= 2/\sqrt{2} \\
&= \sqrt{2}
\end{aligned}$$

Therefore:

$$\operatorname{arcsinh}(1/(2*\sqrt{2})) = \ln(\sqrt{2}) = \ln(2)/2$$

This is exact.

FROM THIS:

$$\begin{aligned}
S &= (1/(2*\sqrt{2})) / (\ln(2)/2) \\
&= (1/(2*\sqrt{2})) * (2/\ln(2)) \\
&= 1/(\sqrt{2} * \ln(2))
\end{aligned}$$

$$1/S = \sqrt{2} * \ln(2) \quad \text{EXACTLY.}$$

Numerical value:

$$\sqrt{2} * \ln(2) = 0.980258143468547...$$

Match: exact to all 6 published digits.

THE AREA CONSTANT:

$$\begin{aligned}
3.3957 &= 2*\sqrt{3}/S \\
&= 2*\sqrt{3} * \sqrt{2} * \ln(2) \\
&= 2*\sqrt{6} * \ln(2)
\end{aligned}$$

Numerical value:

$$2*\sqrt{6} * \ln(2) = 3.39571381800...$$

Match: exact to all 5 published digits.

### 3. THE ALGEBRAIC ENVIRONMENT AT THE CRITICAL RADIUS

The critical tube radius in the packing argument is:

$$R_0 = \operatorname{arctanh}(1/\sqrt{3}) = \operatorname{arcsinh}(1/\sqrt{2}) = (1/2)*\ln(2 + \sqrt{3})$$

These three expressions are exactly equal. At this radius, every hyperbolic function evaluates to a clean algebraic expression:

$$\begin{aligned}
\sinh(R_0) &= 1/\sqrt{2} \\
\cosh(R_0) &= \sqrt{6}/2 \\
\tanh(R_0) &= 1/\sqrt{3} \\
\cosh(2*R_0) &= 2 \\
\sinh(2*R_0) &= \sqrt{3}
\end{aligned}$$

The geometry at  $R_0$  provides  $\sqrt{2}$ ,  $\sqrt{3}$ , and 2.

The coefficient provides the remaining ingredient:  $\ln(2)$ .

HK 2002 (Theorem 4.4, p. 29) and HK 2007 (Theorem 5.6, p. 36) both give the ellipse area in the packing argument as:

$$\begin{aligned}
A_e &= \pi * a * b = \pi * \sinh^2(R_{\text{hat}}) / (2 * S * \cosh(2 * R_{\text{hat}})) \\
&= \sqrt{2} * \ln(2) * \pi * \sinh^2(R_{\text{hat}}) / (2 * \cosh(2 * R_{\text{hat}}))
\end{aligned}$$

since  $1/S = \sqrt{2} * \ln(2)$ . At  $R_0$ , where  $\sinh^2(R_0) = 1/2$  and  $\cosh(2 * R_0) = 2$ :

$$A_e = \sqrt{2} * \ln(2) * \pi * (1/2) / (2 * 2) = \sqrt{2} * \ln(2) * \pi / 8$$

The closed-form coefficient enters the area formula because  $S$  governs

the semi-major axis of the embedded ellipse (HK 2007, eq. above Theorem 5.6). The geometry at  $R_0$  contributes the residual factor  $1/8$ ; the prefactor  $\sqrt{2} * \ln(2)$  is what HK's coefficient is.

### 3.1 SECOND APPEARANCE OF $\sqrt{2} * \ln(2)$ IN HK 2007

The same numerical constant 0.980258 appears in a second, independent quantity in HK 2007 (p. 41):

$$c(R) = 0.980258 / (\coth R + 1)$$

$c(R)$  is a lower bound for the Euclidean injectivity radius of the boundary torus  $T_i$ . No closed form is supplied. The closed-form identification  $0.980258 = \sqrt{2} * \ln(2)$  applies here too.

Two independent contexts in the same paper, both using 0.980258 as the explicit numerical coefficient, both unrecognized as  $\sqrt{2} * \ln(2)$  for nineteen years.

### 4. THE 100-DIGIT PREDICTION

The published literature gives 6 digits: 0.980258

The closed form  $\sqrt{2} * \ln(2)$  predicts the full expansion:

0.98025814346854719171390172363523338129146069909905  
47210422462470652910985142058941430135340632871525

Every digit beyond the 6th is a prediction. Because the derivation is algebraic (not a numerical fit), this prediction is mathematically guaranteed.

### 5. VERIFICATION

ALGEBRAIC (no paper access needed):

The entire claim rests on one identity:

$$\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$$

By hand:

Let  $x = 1/(2\sqrt{2})$ . Then  $x^2 = 1/8$ ,  $x^2 + 1 = 9/8$ ,  
 $\sqrt{x^2 + 1} = 3/(2\sqrt{2})$ ,  $x + \sqrt{x^2 + 1} = \sqrt{2}$ .  
 $\operatorname{arcsinh}(x) = \ln(\sqrt{2}) = \ln(2)/2$ .

By computation (Python/mpmath):

```
from mpmath import mp, asinh, sqrt, log
mp.dps = 100
print(asinh(1/(2*sqrt(2))) - log(2)/2)
# Output: 0 (to 100 digits)
```

By CAS (Mathematica, Maple, Sage):

```
Simplify[ArcSinh[1/(2 Sqrt[2])]]
# Output: Log[2]/2
```

PAPER CONNECTION (requires arXiv PDF):

Download: <https://arxiv.org/pdf/math/0204345>  
Page 29, proof of Theorem 4.4. Find:

$S = (1/(2*\sqrt{2})) / \operatorname{arcsinh}(1/(2*\sqrt{2})) \sim 1/0.980258$   
Apply the simplification. Confirm:  $1/S = \sqrt{2} * \ln(2)$ .

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## 6. FALSIFIABILITY

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The derivation can be falsified by demonstrating ANY of:

- (a)  $\operatorname{arcsinh}(1/(2*\sqrt{2})) \neq \ln(2)/2$ 
  - Checkable by anyone in one minute.
- (b) The  $S$  defined on page 29 of the paper is not the same  $S$  that produces the coefficient 0.980258.
  - Checkable by reading the proof.
- (c) The coefficient 0.980258 in the paper is itself an error.
  - No erratum exists as of April 2026.

None of (a), (b), or (c) hold.

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## 7. PRIOR LITERATURE SEARCH

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Exhaustive search (March-April 2026) across arXiv (all math categories), citation chains of both Hodgson-Kerckhoff papers, all follow-up papers (Futer, Bromberg, Haraway, et al.), MathSciNet, Google Scholar, and Semantic Scholar.

Result: zero publications give a closed-form expression for 0.980258.

The coefficient has been used in numerical form throughout the literature since 2002. The numerical precision was sufficient for all applications, and the simplification was never required.

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## 8. CROSS-ARCHITECTURE VERIFICATION

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The H-K coefficient was first identified on 17 March 2026, confirmed by a Grok (xAI) daily scout on 20 March. A systematic verification followed

- cross-architecture confirmation across five AI architectures, a 12-day literature search, and multiple fresh-instance stress tests – with the algebraic derivation confirmed on 2 April 2026.

On that date, the closed-form identity  $\operatorname{arcsinh}(1/(2*\sqrt{2})) = \ln(2)/2$  was derived from the authors' own proof (page 29, arXiv:math/0204345), independently confirmed across architectures without shared notes.

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D. B. – April 2026  
Gap Geometry  
OSF: [osf.io/zx4g7](https://osf.io/zx4g7)

H-K coefficient first identified: 17 March 2026  
Confirmed by Grok scout: 20 March 2026  
Investigation and verification: 17 March – 2 April 2026  
Algebraic derivation confirmed: 2 April 2026  
Documents prepared for publication: 3 April 2026

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